

CSCI E-102 Econometrics and Causal Inference with R

Harvard Extension School

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Lecture 5

1 Econometric Models: Linear Regression (Continued)

- Testing Hypotheses and Confidence Interval for Slope
 - Approximate Distribution of Test Statistic
 - Testing Hypotheses about Slope β_1
 - Confidence Interval for β_1
- Testing Hypotheses and Confidence Interval for Intercept
 - Approximate Distribution of Test Statistic
 - Testing Hypotheses about Intercept β_0
 - Confidence Interval for β_0
- Regression Model with Dummy Variables
- Homoskedasticity vs. Heteroskedasticity
 - Definition
 - Homoskedasticity-only Standard Error
- Best Linear Unbiased Estimator (BLUE)

2 Multivariate Linear Regression Model

- Terminology and Assumptions of the Multiple Regression

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2 Multivariate Linear Regression Model

- Terminology and Assumptions of the Multiple Regression

Approximate Distribution of Test Statistic

Recall:

For a Linear Regression Econometric Model, the OLS estimator

$$\hat{\beta}_1 \xrightarrow{D} N(\beta_1, \sigma_{\hat{\beta}_1}^2) \text{ as } n \rightarrow \infty, \text{ where } \sigma_{\hat{\beta}_1}^2 = \frac{1}{n} \frac{\text{Var}((X_i - \mu_X)u_i)}{(\text{Var}(X_i))^2}.$$

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Claim:

Given Econometric model,

$$Y_i = \beta_0 + \beta_1 X_i + u_i \text{ with } i \in \{1, 2, \dots, n\},$$

as $n \rightarrow \infty$, we have:

$$\frac{\hat{\beta}_1 - \beta_1}{\hat{\sigma}_{\hat{\beta}_1}} \overset{A}{\sim} t_{n-2}, \text{ where } \hat{\sigma}_{\hat{\beta}_1}^2 = \frac{1}{n} \frac{\frac{1}{n-2} \sum_{i=1}^n (X_i - \bar{X})^2 \hat{u}_i^2}{\left(\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2\right)^2}.$$

Here, $\overset{A}{\sim}$ denotes an approximate distribution (i.e. they converge in distribution).

Heteroskedasticity-robust Standard Error of $\hat{\beta}_1$

Terminology:

The standard error of $\hat{\beta}_1$ given by

$$\hat{\sigma}_{\hat{\beta}_1}^2 = \frac{1}{n} \frac{\frac{1}{n-2} \sum_{i=1}^n (X_i - \bar{X})^2 \hat{u}_i^2}{\left(\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \right)^2}$$

is called *heteroskedasticity-robust standard error*.

Testing Hypotheses about Slope β_1

Corollary:

Given large n , in order to test

$$H_0 : \beta_1 = \beta_{1,0} \text{ vs. } H_1 : \beta_1 \neq \beta_{1,0},$$

one can look at the following test statistic:

$$\begin{aligned} \text{TS} &\doteq \frac{\hat{\beta}_1 - \beta_{1,0}}{\hat{\sigma}_{\hat{\beta}_1}} \\ &= \frac{\hat{\beta}_1 - \beta_{1,0}}{\sqrt{\frac{1}{n} \frac{\sum_{i=1}^n (X_i - \bar{X})^2 \hat{u}_i^2}{\sum_{i=1}^n (X_i - \bar{X})^2}}}, \end{aligned}$$

which approximately follows t_{n-2} under $H_0 : \beta_1 = \beta_{1,0}$.

Note:

Similarly, using this test statistic, TS, one can conduct one-sided tests.

Confidence Interval for β_1

Corollary:

For large n , $(1 - \alpha)100\%$ confidence interval for β_1 is given by:

$$\begin{aligned}\beta_1 &= \hat{\beta}_1 \pm t_{n-2, \alpha/2} \hat{\sigma}_{\hat{\beta}_1} \\ &= \hat{\beta}_1 \pm t_{n-2, \alpha/2} \sqrt{\frac{1}{n} \frac{\frac{1}{n-2} \sum_{i=1}^n (X_i - \bar{X})^2 \hat{u}_i^2}{\left(\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2\right)^2}}.\end{aligned}$$

Here, $t_{n-2, \alpha/2}$ denotes a number such that $F_{t_{n-2}}(t_{n-2, \alpha/2}) = 1 - \alpha/2$, where $F_{t_{n-2}}(\cdot)$ denotes the cdf of the t_{n-2} distribution.

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$$\hat{\beta}_0 \xrightarrow{D} N(\beta_0, \sigma_{\hat{\beta}_0}^2) \text{ as } n \rightarrow \infty, \text{ where } \sigma_{\hat{\beta}_0}^2 = \frac{1}{n} \frac{\text{Var}(H_i u_i)}{(\text{E}(H_i^2))^2}, \quad H_i = 1 - \left[\frac{\mu_X}{\text{E}[X_i^2]} \right] X_i.$$

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Claim:

Given Econometric model,

$$Y_i = \beta_0 + \beta_1 X_i + u_i \quad \text{with } i \in \{1, 2, \dots, n\},$$

as $n \rightarrow \infty$, we have:

$$\frac{\hat{\beta}_0 - \beta_0}{\hat{\sigma}_{\hat{\beta}_0}} \overset{A}{\sim} t_{n-2}, \text{ where } \hat{\sigma}_{\hat{\beta}_0}^2 = \frac{1}{n} \frac{\frac{1}{n-2} \sum_{i=1}^n \hat{H}_i^2 \hat{u}_i^2}{\left(\frac{1}{n} \sum_{i=1}^n \hat{H}_i^2 \right)^2}, \quad \hat{H}_i = 1 - \left[\frac{\bar{X}}{\frac{1}{n} \sum_{i=1}^n X_i^2} \right] X_i.$$

Here, $\overset{A}{\sim}$ denotes an approximate distribution (i.e. they converge in distribution).

Heteroskedasticity-robust Standard Error of $\hat{\beta}_0$

Terminology:

The standard error of $\hat{\beta}_0$ given by

$$\hat{\sigma}_{\hat{\beta}_0}^2 = \frac{1}{n} \frac{\frac{1}{n-2} \sum_{i=1}^n \hat{H}_i^2 \hat{u}_i^2}{\left(\frac{1}{n} \sum_{i=1}^n \hat{H}_i^2 \right)^2}, \quad \text{where } \hat{H}_i = 1 - \left[\frac{\bar{X}}{\frac{1}{n} \sum_{i=1}^n X_i^2} \right] X_i,$$

is called *heteroskedasticity-robust standard error*.

Testing Hypotheses about Intercept β_0

Corollary:

Given large n , in order to test

$$H_0 : \beta_0 = \beta_{0,0} \text{ vs. } H_1 : \beta_0 \neq \beta_{0,0},$$

one can look at the following test statistic:

$$TS \doteq \frac{\hat{\beta}_0 - \beta_{0,0}}{\hat{\sigma}_{\hat{\beta}_0}},$$

where

$$\hat{\sigma}_{\hat{\beta}_0}^2 = \frac{1}{n} \frac{\frac{1}{n-2} \sum_{i=1}^n \hat{H}_i^2 \hat{u}_i^2}{\left(\frac{1}{n} \sum_{i=1}^n \hat{H}_i^2 \right)^2}, \quad \hat{H}_i = 1 - \left[\frac{\bar{X}}{\frac{1}{n} \sum_{i=1}^n X_i^2} \right] X_i,$$

which approximately follows t_{n-2} under $H_0 : \beta_0 = \beta_{0,0}$.

Note:

Similarly, one can conduct one-sided tests.

Confidence Interval for β_0

Corollary:

For large n , $(1 - \alpha)100\%$ confidence interval for β_0 is given by:

$$\beta_0 = \hat{\beta}_0 \pm t_{n-2, \alpha/2} \hat{\sigma}_{\hat{\beta}_0},$$

where

$$\hat{\sigma}_{\hat{\beta}_0}^2 = \frac{1}{n} \frac{\frac{1}{n-2} \sum_{i=1}^n \hat{H}_i^2 \hat{u}_i^2}{\left(\frac{1}{n} \sum_{i=1}^n \hat{H}_i^2 \right)^2}, \quad \hat{H}_i = 1 - \left[\frac{\bar{X}}{\frac{1}{n} \sum_{i=1}^n X_i^2} \right] X_i,$$

Here, $t_{n-2, \alpha/2}$ denotes a number such that $F_{t_{n-2}}(t_{n-2, \alpha/2}) = 1 - \alpha/2$, where $F_{t_{n-2}}(\cdot)$ denotes the cdf of the t_{n-2} distribution.

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Regression Model with Dummy Variables

Let's consider the following Econometric Model

$$Y_i = \beta_0 + \beta_1 D_i + u_i \quad \text{with } i \in \{1, 2, \dots, n\},$$

where the regressor is *binary* (or *dummy*): $D_i \in \{0, 1\}$.

Then

$$Y_i = \beta_0 + u_i, \text{ if } D_i = 0 \text{ and}$$

$$Y_i = \beta_0 + \beta_1 + u_i, \text{ if } D_i = 1$$

describe two populations with means

$$E[Y_i | D_i = 0] = \beta_0 \text{ and}$$

$$E[Y_i | D_i = 1] = \beta_0 + \beta_1, \text{ respectively.}$$

Therefore, the slope

$$\beta_1 = E[Y_i | D_i = 1] - E[Y_i | D_i = 0]$$

is the difference between the population means.

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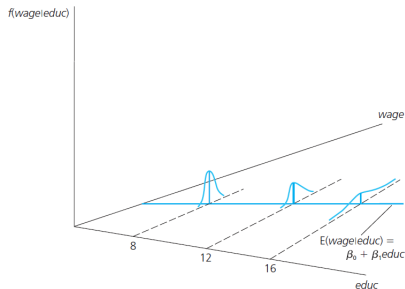
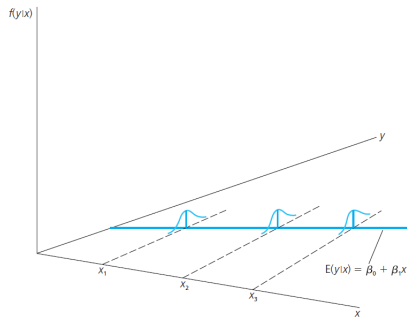
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Homoskedasticity vs. Heteroskedasticity

Example:

Homoskedastic (left) and heteroskedastic (right) errors.



Source: *Introductory Econometrics: A Modern Approach* by Jeffrey Wooldridge

Homoskedasticity vs. Heteroskedasticity

Def.

Given Econometric model,

$$Y_i = \beta_0 + \beta_1 X_i + u_i \quad \text{with } i \in \{1, 2, \dots, n\},$$

the error term u_i is called

- *homoskedastic* if $\text{Var}(u_i | X_i = x)$ is constant w.r.t. i and x
- *heteroskedastic* if it is not homoskedastic

Homoskedasticity-only Standard Error

Claim:

Given Econometric model,

$$Y_i = \beta_0 + \beta_1 X_i + u_i \quad \text{with } i \in \{1, 2, \dots, n\},$$

with homoskedastic error term u_i , the estimators of

$$\text{Var}(\hat{\beta}_0) \quad \text{and} \quad \text{Var}(\hat{\beta}_1)$$

are given by the following *homoskedasticity-only* estimators:

$$\tilde{\sigma}_{\hat{\beta}_0}^2 = \frac{s_{\hat{u}}^2 \sum_{i=1}^n X_i^2}{n \sum_{i=1}^n (X_i - \bar{X})^2} \quad \text{and} \quad \tilde{\sigma}_{\hat{\beta}_1}^2 = \frac{s_{\hat{u}}^2}{\sum_{i=1}^n (X_i - \bar{X})^2},$$

where

$$s_{\hat{u}}^2 = \frac{\text{SSR}}{n-2} = \frac{\sum_{i=1}^n (Y_i - \hat{Y}_i)^2}{n-2}.$$

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Best Linear Unbiased Estimator (BLUE)

Def.

Given Econometric model,

$$Y_i = \beta_0 + \beta_1 X_i + u_i \quad \text{with } i \in \{1, 2, \dots, n\},$$

the estimator

$$\tilde{\beta}_1 \doteq \sum_{i=1}^n a_i Y_i,$$

where $a_1, a_2, \dots, a_n$ can depend on $X_1, X_2, \dots, X_n$, is called *linear* estimator of β_1 .

If also

$$E[\tilde{\beta}_1 | X_1, \dots, X_n] = \beta_1,$$

the estimator $\tilde{\beta}_1$ is called *linear conditionally unbiased* estimator.

Best Linear Unbiased Estimator (BLUE)

Theorem (Gauss-Markov Theorem)

Given Econometric model,

$$Y_i = \beta_0 + \beta_1 X_i + u_i \quad \text{with } i \in \{1, 2, \dots, n\},$$

where

- 1 *exogeneity* assumption: $E[u_i | X_i] = 0$
- 2 (X_i, Y_i) is i.i.d. sample drawn from a joint distribution
- 3 $E[X_i^4]$ and $E[Y_i^4]$ are nonzero and finite

and the error term u_i is homoskedastic then the OLS estimator

$\hat{\beta}_1$ is the **B**est (most efficient) **L**inear conditionally **U**nbiased **E**stimator (**BLUE**).

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Multivariate Linear Regression: Terminology and Assumptions

Terminology:

We assume that $X_{1i}, X_{2i}, \dots, X_{ki}$ and Y_i are random and there is relation

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_1 X_{ki} + \dots + \beta_k X_{ki} + u_i,$$

where

- index $i \in \{1, 2, \dots, n\}$
- Y_i is called *dependent variable* or *response*
- $X_{1i}, X_{2k}, \dots, X_{ki}$ are the i^{th} observations on each of the k regressors
- function $\beta_0 + \beta_1 X_{1i} + \beta_1 X_{ki} + \dots + \beta_k X_{ki}$ is referred as *population regression line*
- β_0 is *intercept* of the population regression line
- $\beta_1, \beta_2, \dots, \beta_k$ is *slope coefficient* on $X_{1i}, X_{2k}, \dots, X_{ki}$, respectively
- u_i is the *error term*

Multivariate Linear Regression: Terminology and Assumptions

Assumptions:

We assume that there is the following relation between X_i and Y_i :

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_k X_{ki} + u_i \quad \text{with } i \in \{1, 2, \dots, n\},$$

where

- ① *exogeneity* assumption: $E[u_i | X_{1i}, X_{2i}, \dots, X_{ki}] = 0$
- ② $(X_{1i}, X_{2i}, \dots, X_{ki}, Y_i)$ is i.i.d. sample drawn from a joint distribution
- ③ $E[X_{1i}^4], E[X_{2i}^4], \dots, E[X_{ki}^4]$ and $E[Y_i^4]$ are nonzero and finite
- ④ there is no perfect *multicollinearity*: none of the regressors is a linear function of the other regressors